```python
from itertools import permutations

def travelling_salesman(graph, start):
    cities = list(graph.keys())
    cities.remove(start)

    shortest_distance = float('inf')
    best_route = None

    for route in permutations(cities):
        current_city = start
        distance = 0

        for next_city in route:
            distance += graph[current_city][next_city]
            current_city = next_city

        # Return to starting city
        distance += graph[current_city][start]

        if distance < shortest_distance:
            shortest_distance = distance
            best_route = (start,) + route + (start,)

    return best_route, shortest_distance


graph = {
    'A': {'B': 10, 'C': 15, 'D': 20},
    'B': {'A': 10, 'C': 35, 'D': 25},
    'C': {'A': 15, 'B': 35, 'D': 30},
    'D': {'A': 20, 'B': 25, 'C': 30}
}

route, distance = travelling_salesman(graph, 'A')

print("Shortest Route:", " -> ".join(route))
print("Minimum Distance:", distance)
```

IDLE Shell 3.10.0

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```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=============== RESTART: C:/Users/hi/OneDrive/Desktop/AI/expp.py ===============
Shortest Route: A -> B -> D -> C -> A
Minimum Distance: 80
>>>
```

Ln: 7 Col: 0

expp.py - C:/Users/hi/OneDrive/Desktop/AI/expp.py (3.10.0)

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route, distance = travelling_salesman(graph, 'A')

print("Shortest Route:", " -> ".join(route))
print("Minimum Distance:", distance)
```

Ln: 38 Col: 36